

CLIMATE-FACT-SHEET

Mafia Island Marine Park (MIMP), Tanzania

Supporting climate-smart MPA management in the Western Indian Ocean



Abstract

Marine Protected Areas in the Western Indian Ocean are on the frontline of climate change. Rising sea temperatures, shifting ocean chemistry, accelerating sea level rise, and increasingly erratic rainfall are fundamentally altering the ecosystems that MPAs are designed to protect and the livelihoods of the communities that depend on them. Effective MPA management increasingly requires climate literacy: the ability to understand, interpret, and act on climate data.

Mafia Island Marine Park occupies a particularly exposed position within the Western Indian Ocean region. Located within the East African Coastal Current system, MIMP's reefs, seagrass meadows, and mangrove forests are subject to the compound pressures of warming, acidification, and sea-level rise, layered on top of longstanding local stressors, including fishing pressure, sedimentation from the Rufiji River delta, and land-based pollution. The communities that depend on MIMP, approximately 66,180 people on Mafia Island alone, have limited adaptive capacity and few formal safety nets to buffer the economic shocks that ecological degradation brings.

Despite this exposure, climate change remains inadequately integrated into MPA management due to a lack of marine climate change indicators across the Western Indian Ocean region. Management plans are largely built on historical baselines that no longer reflect current or near-future conditions. Decision-making frameworks rarely incorporate quantified climate thresholds, and the gap between available critical climate data and information and park managers' and rangers' practical knowledge remains wide. The ClimPA initiative exists specifically to bridge that gap.

This factsheet translates available essential climate variable into decision-relevant information for the managers responsible for Mafia Island Marine Park (MIMP). It is organised around the eight climate variables most relevant to MIMP management, each presented with a clear narrative from historical trends through to future projections under contrasting emission scenarios. It is a living document, designed to be updated as local monitoring data and refined projections become available, and it is intended to anchor the climate change section of MIMP's next General Management Plan revision..



2. Current Climate & Historical Trends

Sea surface temperature

+0.16°C/decade since 1985 (0.6°C total).

Observational records confirm significant baseline warming across MIMP waters, with cooler months showing an accelerated warming trend by 25% relative to historical averages. This persistent thermal upward heightens baseline stress on coastal ecosystems, priming habitats for severe mass mortality during anomalous regional warming periods such as the 2016 and 2020 coral bleaching events.

Ocean acidification

pH 8.08 historical baseline 2000–2018.

The historical mean pH baseline of 8.08 marks the reference point against which future ocean acidification will be measured. As atmospheric CO₂ continues to accumulate, the resulting decline in carbonate ion availability threatens the calcification capacity of reef-building corals and other calcifying organisms.

Sea level rise

+3.47mm/yr, 1993–2025 (10.7cm cumulative).

Satellite altimetry records combined with regional tide gauge verification at Zanzibar establish a steady, accelerating sea level rise trend of nearly 3.5 mm per year over the past three decades. This compounding vertical rise progressively accelerates shoreline erosion, exacerbates saltwater intrusion into coastal freshwater lenses, and starves landward ecosystems of natural migration space.

Rainfall pattern

Bimodal Masika long rains (Mar–May), Vuli short rains (Oct–Dec)

Rainfall is highly seasonal, defined by a distinct peak during the long rains in April (244mm) and March (172mm), followed by a secondary peak in December (112mm). The dry season spans from June through October, with July experiencing the lowest rainfall (11mm). Mean surface air temperatures remain warm year-round, ranging from a cooler period around July (23.95°C mean / 19.38°C min) to peak warmth in February and March (28.04°C and 27.99°C mean).

Marine heat waves

150+ days/yr since 2015 (few/none in 1980s–90s).

Marine heatwaves have undergone a structural shift, transitioning from rare anomalies in the late 20th century to a chronic, near-annual phenomenon impacting MIMP. The record-breaking 2024 event established a new record of prolonged thermal stress lasting over 150 days, stripping benthic habitats of normal recovery windows.

Salinity

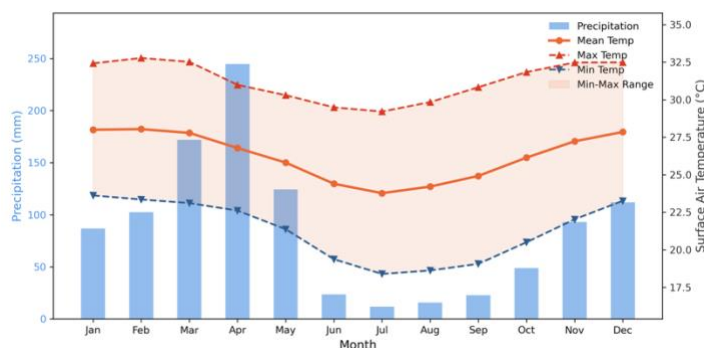
33.61psu historical baseline 2000–2019 mean.

The historical mean salinity baseline of 33.61 PSU serves as a placeholder for local hydrodynamic conditions, which are heavily influenced by freshwater pulses from the Rufiji River delta.

Storms & cyclone exposure

3 record-intensity systems since 2014.

Although traditionally outside the primary cyclone belt, MIMP has experienced a notable shift in storm exposure frequency and intensity over the past decade, highlighted by systems such as Hellen (2014), Freddy (2023), and Hidaya (2024) which made landfall. These extreme events generate high-energy wave action that physically fractures fragile branching coral structures and strips coastal protection buffers.

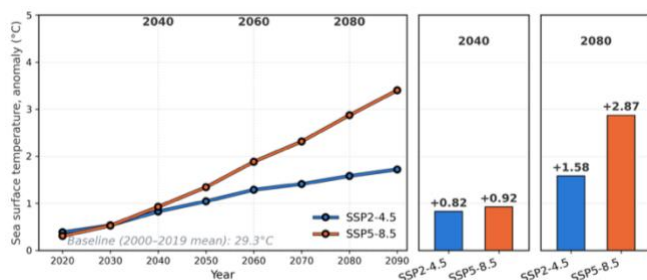


3 Summary of Projected Future Climate

This section provides an overview of projected possible changes for selected marine climate parameters at MIMP. It is generated from the best currently available data; it does not replace site-specific impact or adaptation studies. As such, the information presented here does not contain information on the finer local scale changes that may occur, and which may be of more interest. Each row below reflects the middle-of-the-road (SSP2-4.5) and high-emissions (SSP5-8.5) pathways, to 2100.

Sea surface temperature

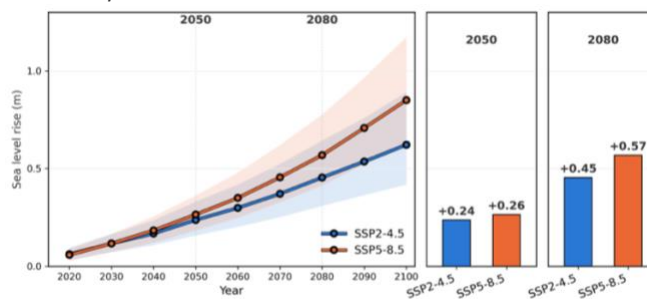
Mid-century warming of +1.0°C above baseline is effectively locked in under both pathways. By end-century, warming levels off near +1.5°C under SSP2-4.5 versus +3.0°C under SSP5-8.5.



Driven by rising sea surface temperatures, regional MHW hotspots will transition toward a near-permanent state by mid-century (2040–2050), with extreme Category IV events rapidly increasing by 2060. By end-century, durations are projected to surge to 300+ days annually, with intensity reaching +1.5°C under SSP2-4.5 and exceeding +2.0°C under the high-emissions pathway (SSP5-8.5).

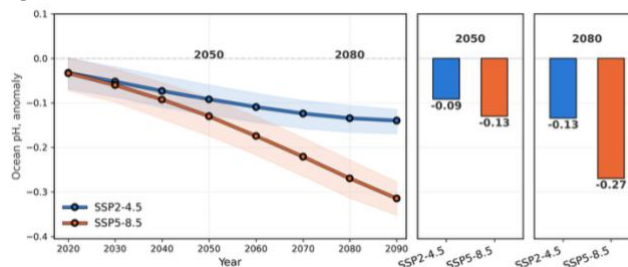
Sea level rise

The likely range of projected change in mean sea level rise near Mafia Island Marine Park is from +0.42 to +0.89 m for the low emission scenario and from +0.61 to +1.17 m for the high emission scenario by 2100.



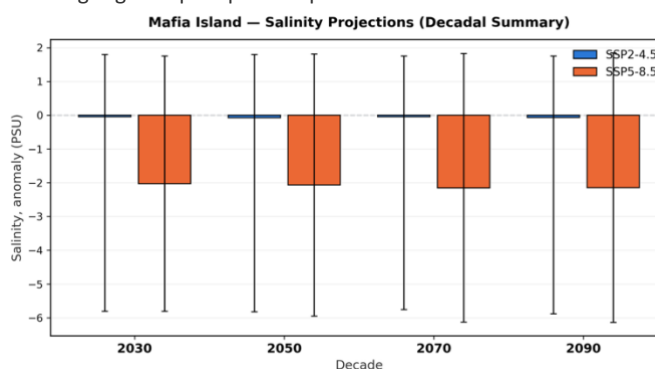
Ocean acidification

Projections indicate a progressive decline in surface seawater pH across MIMP. Under the intermediate scenario (SSP2-4.5), pH anomaly reaches -0.09 by 2050 and -0.13 by end-century. Under the high-emissions pathway (SSP5-8.5), acidification accelerates sharply, projecting a pH anomaly of -0.13 by 2050 and plunging to -0.27 by 2080–2090. This chemical shift is projected to drastically reduce aragonite and calcite saturation states, severely impeding calcification rates in reef-building corals and crustose coralline algae.



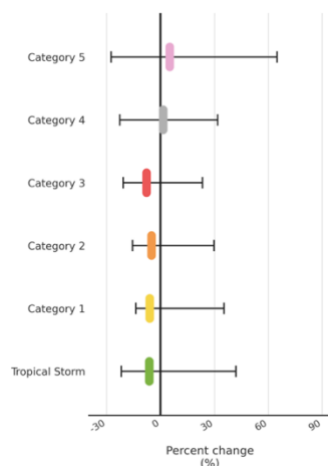
Salinity

Decadal salinity anomaly projections indicate considerable inter-annual variability spanning roughly -6 to +2 PSU around baseline conditions. While median central anomalies remain close to 0 PSU under SSP2-4.5, high-emission scenarios (SSP5-8.5) suggest a persistent slight negative salinity anomaly of approximately -2 PSU across decades, driven by intensified freshwater runoff pulses and shifting regional precipitation patterns.



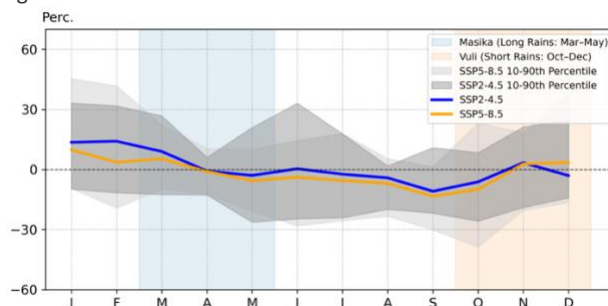
Storms & extreme events

Projections show a category-dependent pattern rather than a uniform trend: weaker systems (tropical storm through Category 3) show median frequency declines of roughly -10% across the Tanzanian EEZ, while Category 5 landfalls show a median increase of +5.2% a "fewer but more intense" signal. Confidence is low



Rainfall & freshwater runoff

Seasonal projections indicate this upward trend with rainfall increasing by a median of +3% to +15% across SSP2-4.5 and SSP5-8.5. These shifts toward concentrated, intense rainfall events drive high-volume terrestrial runoff



4 Impacts, Risk & Vulnerability

Risk is understood as the product of hazard exposure, ecological/social sensitivity, and adaptive capacity, applied here alongside each system's projected impact within its own card, so cause and consequence sit together rather than needing two separate lookups. Compound risks, where multiple hazards interact, are noted below the ecological cards. Each impact line carries a citation back to the Section 3 figure it is derived from.

Ecological systems

Coral reefs.

Very high

Warming seas and increasingly frequent marine heatwaves are pushing conditions beyond coral thermal tolerance, making annual bleaching events likely to become the norm. Compounding this, ocean acidification weakens the calcification corals need to rebuild and recover between stress events, leaving the reef system with the least capacity to bounce back of any ecosystem in the park.

Exposure: very high Sensitivity: very high
Adaptive capacity: low

Seagrass meadows.

Very high

Rising sea temperatures increase the likelihood of thermal stress episodes for seagrass meadows, raising the risk of periodic die-off. Because recovery depends on water clarity, which is itself vulnerable to erosion and sedimentation, repeated stress events risk compounding into a longer-term decline in meadow extent.

Exposure: high Sensitivity: high Adaptive capacity: low

Mangrove forests.

High

Rising sea levels raise the likelihood of widespread inundation in mangrove stands that lack space to migrate landward. Under sustained waterlogging, peat oxidation converts these areas from carbon sinks into carbon sources, while their nursery function for reef and inshore fish species is likely to degrade.

Exposure: high Sensitivity: medium
Adaptive capacity: low-medium

Sea weed.

Very high

Sustained ocean warming pushes cultivated seaweed beyond its thermal tolerance with increasing frequency, raising the likelihood of recurring crop stress and die-back. Compounding this, declining nearshore salinity from increased rainfall and Rufiji River freshwater discharge adds osmotic stress on top of the thermal pressure, weakening the plant's overall resilience.

Exposure: high Sensitivity: high Adaptive capacity: low

Socioeconomic sectors

Ecological degradation at MIMP translates directly into livelihood losses for some of the most economically vulnerable communities. Compound losses across reef fisheries, seagrass-associated species, and intertidal foraging will reduce protein availability for a population already dependent on marine resources for nutritional security.

Artisanal fisheries. High Continued warming and more frequent bleaching events are likely to drive a substantial decline in reef fish biomass, reducing catch income for fishing households across MIMP. With no formal safety net or alternative income buffer currently in place, this loss would fall directly on communities least able to absorb it.	Seaweed farming. Medium-high Sustained ocean warming raises the risk of seaweed farming failure as water temperatures push beyond the crop's thermal tolerance. Declining nearshore salinity from increased rainfall and Rufiji River freshwater discharge compounds the stress. Together, these pressures raise income volatility for farming households, with no climate-resilient cultivar substitution programme currently in place to buffer the exposure.
Tourism. Medium Reef degradation will reduce MIMP's international draw for diving and snorkelling, threatening ~120 enterprises; beach erosion from SLR may affect coastal resort infrastructure by 2050 under higher-emission scenarios.	Coastal settlements. High Accelerated sea-level rise (+0.61 to +1.17m under SSP5-8.5 by 2100) combined with intensified Category 5 storm surges presents a near-certain hazard probability for low-lying residential areas, threatening permanent displacement, saltwater contamination of community freshwater wells, and structural damage to public infrastructure

5 Implications for MPA Management

Climate-smart zoning

Current MIMP zoning does not account for climate change impacts. Effort should be put in place to identify and strictly protect climate refugia reef sites with natural thermal buffering and establish adaptive management triggers that activate enhanced protection automatically. Landward migration corridors for mangroves should be protected.

Monitoring priorities

Priority monitoring investments: (1) SST loggers; (2) annual GCRMN-protocol coral transect surveys; (3) ecosystem mapping; (4) community-based catch monitoring aligned to TAFIRI stock assessments. All data to be stored in FAIR-compliant format.

Early-warning systems

MIMP requires a tiered bleaching early-warning system: (1) NOAA CoralWatch DHW alert subscribed for MIMP coordinates; (2) Data logger installed should be retrieved and data analysed; (3) bleaching response protocol activated when DHW exceeds 4 °C-weeks (watch) and 8 °C-weeks (warning). SLR early-warning should monitor tide gauge anomalies and issue pre-storm surge alerts for coastal settlements.

Integration into management plans

The MIMP General Management Plan (next revision) must include: (a) a climate change section; (b) quantified thresholds that trigger management reviews; (c) an adaptive management timeline aligned to this factsheet; (d) a climate finance strategy referencing Green Climate Fund eligibility; (e) annual data review requirement.

How This Factsheet Was Developed

The factsheet was developed iteratively through the described approaches.

1. Stakeholder Engagement

Variable selection and the framing of management thresholds followed a human-centered design (HCD) approach: rather than starting from what data is available and working forward, the process started from what MIMP-facing stakeholders need to act on, and worked backward to the variables, thresholds, and framing that make the factsheet usable in an actual management decision.

This was carried out through two distinct engagement tracks:

- **Stakeholder meetings**, bringing together MIMP-facing groups to surface management priorities, existing gaps in current planning processes, and the practical decisions zoning, fisheries regulation, infrastructure siting, bleaching response that climate information needs to support.
- **Climate technical expert meetings**, validating the scientific framing, data sources, and threshold definitions against current climate science and regional data availability, and checking that the language used would hold up to technical scrutiny without losing accessibility for non-specialist managers.

Together, these two tracks shaped three concrete elements of the factsheet: the five variables ultimately selected (Section 2, Collating Essential Climate Variables), the Past → Present → Future narrative structure used for each variable card, and the specific management thresholds highlighted throughout the document.

2. Collating Essential Climate Variables (ECVs)

Five variables were selected for their direct relevance to MPA management decisions (zoning, fisheries regulation, infrastructure siting), aligned with the GCOS Essential Climate Variable framework applied to a coastal/marine context: sea surface temperature, ocean acidification, sea level rise, storms/extreme events, and rainfall/freshwater runoff. Each maps to an identifiable management lever SST to bleaching-response protocols, rainfall to Rufiji delta turbidity management, SLR to coastal infrastructure setback planning.

3. Data Processing

Each variable draws on a distinct source and processing method, summarised in the Data Pathway table on the preceding page: satellite-derived SST and marine heatwave metrics (NOAA Coral Reef Watch); ocean acidification from IPCC AR6 WGII global reference rates, pending in-situ MIMP measurement; sea level from satellite altimetry linear-trend fitting for the historical record and CMIP6 ensemble percentile extraction for future scenarios; storm frequency from CHEZ; and rainfall from CRU historical time series combined with seasonal SSP2-4.5/SSP5-8.5 percent-change projections by month.

4. Archiving

Processed datasets and intermediate outputs are held in an institutional repository with version control, rather than in ad hoc local files. This means each revision made to this factsheet (CHEZ storm data, CMIP6 sea level, seasonal rainfall) can in principle be traced back to a specific, reproducible processing run rather than a static number pasted into a document. A remaining item is to confirm whether the repository has a defined refresh cadence tied to this factsheet's stated revision date, and who is responsible for triggering re-processing when new source data lands.

